

Dhrubajit Paul Chowdhury

Senior Full-Stack Software Engineer — React · TypeScript · Node.js · Python

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EXECUTIVE SUMMARY

Full-stack engineer, 9+ years, building React/TypeScript frontends and the services behind them (Node.js, Python) across fintech, investment, loyalty and transport. Takes features from technical design through to production readiness: API design, authentication, observability and layered test automation. Works AI-assisted (Claude Code, GitHub Copilot, spec-driven development) to ramp into unfamiliar codebases fast, shipping production features on a merged banking platform in week one. Leads small teams, mentors engineers on testing and craftsmanship, speaks at regional developer events.

KEY COMPETENCIES

Frontend: React · TypeScript · Next.js · Module Federation (Vite) · React Query · Angular · NgRx

AI Engineering: Claude Code · GitHub Copilot · Cursor · Spec-driven development · Agent skills · Agentic RAG

Backend & APIs: Kotlin · Spring Boot · Node.js · Express · Python (FastAPI, Flask) · Ruby on Rails · REST API design · GraphQL · Microservices · PostgreSQL · MSSQL

Testing & Quality: Cypress · Playwright · BDD/TDD · E2E automation strategy · Service virtualisation (WireMock, LocalStack)

Ops & Observability: Docker / Docker Compose · CI/CD · Grafana · Loki · Promtail · Keycloak · AWS (EC2) · Azure Cloud

EXPERIENCE

Senior Software Engineer | Zühlke Engineering, Singapore

Aug 2025 – Present

Isolated local development environment for a legacy .NET microservices platform, GIC

Developers used to compete for a single shared UAT environment to validate any change. Delivered an environment that runs entirely on a developer laptop, with no VPN and no corporate dependencies, taking the feedback cycle from roughly a week to minutes.

- Built out the internal developer tooling: a one-command Node.js CLI that provisions and orchestrates the full multi-service stack.
- Designed the Docker and Docker Compose infrastructure, standing up WireMock and LocalStack in place of the services that can't run locally, so the environment needs no shared staging estate.
- Adapted the Cypress E2E suites to the local stack, making end-to-end testing reliable before code reaches a shared environment.
- Kept it working across Windows and Linux, and left a framework for adding further business workflows.
- AI-augmented: used AI to research and document the legacy .NET codebase, reconstructing dependencies, API flows and seed data, to deliver inside the project's time constraints.

Automated client information management and compliance monitoring, UBS

Backend engineering on a client-onboarding initiative after two banking platforms merged: track changes in client circumstances, trigger alerts to update client information, monitor expiry of the documents required to maintain tax status.

- Migrated features across the two merged banking systems so tax workflows integrated cleanly, and enabled recalculation of tax-related fields without breaking regulatory compliance functions.
- Extended the system to track client circumstance changes and fire document-expiry alerts, which gave compliance better oversight.
- Led the React/TypeScript micro-frontend migration from Create React App Module Federation to Vite Module Federation, cutting build times and tightening the local feedback loop.
- AI-augmented: reconstructed the business workflows, compliance rules and existing behaviour of an unfamiliar platform through AI-assisted exploration, and ramped on the existing patterns fast enough to ship production features in the first week.

Lead Software Engineer (Contract) | Alstom, Singapore

May 2025 – Jul 2025

Centralised defect capture and triage platform, rail maintenance

No centralised defect database existed, so nobody could reference past defects for comparison. Built a mobile capture app for maintenance crews and a triage catalogue for supervisors.

- Led a team of 3 to deliver APAC's first centralised railway defect capture and catalogue system, from requirements to production in 3 months and approved for further investment.
- Built two frontends on one workflow: a mobile capture app for field crews, and an internal console where supervisors triage a defect against historical cases.
- Owned production readiness: Keycloak authentication and authorisation, plus a Grafana/Loki/Promtail logging and monitoring stack that removed the guesswork from debugging and sped up resolution of user-reported issues.
- Integrated with a data engineer's image-processing pipeline to push real-time notifications into operators' LINE app.
- Built an offline train inspection flow on a Raspberry Pi capture unit and iPad client, cutting manual inspection from 3 days to about 4 hours.
- AI-augmented: used AI to compress the build time on both the auth layer and the observability stack, which is how a 3-month scope absorbed them.

Senior Full-Stack Software Engineer | Ascenda Loyalty, Singapore

Feb 2022 – May 2025

Loyalty-as-a-Service platform: consumer redemption for bank and fintech partners

SaaS web application where end users redeem points accrued through banks and credit-card partners, browse redemption options, and set up auto-redemption. MVP launched in three months, then enhanced continuously.

- Led a team of 3 through rapid iterative prototyping to launch the flagship MVP in three months.
- Designed the checkout flow across multiple redemption types, and built the interaction components with the design team, making sure they met accessibility requirements.
- Designed and built a self-service platform in Next.js so enterprise clients could configure and deploy their own branded redemption experiences.
- Drove a company-wide automated E2E testing strategy with QA, raising deployment confidence. Bugs escaping manual testing fell ~40%.

Software Engineering Consultant | Zenika, Singapore

Jul 2018 – Jan 2022

Global asset allocation dashboard, GIC

Dashboard monitoring global equity and commodity movements, supporting opportunistic asset allocation off daily price movements and analytics.

- Built RESTful APIs in Python/FastAPI to extract and process asset data from MSSQL.
- Developed interactive market visualisations and commodity charts (ECharts, AgGrid, Ant Design) in React/TypeScript, giving analysts access to critical investment data about 50% faster than the previous system.
- Containerised the application stack with Docker and deployed it to EC2.

Other client engagements

- Reflaunt: trained and mentored 10+ engineers on BDD/TDD and software craftsmanship; post-deployment defects fell 25%.
- Decathlon: designed a micro-frontend strategy to unify disparate vendor applications, with a roadmap projected to save 30% on future shared-component development.
- GovTech: made the case for moving the MyCareersFuture job portal off GraphQL to REST and started the migration, for better API performance and simpler integrations.
- AXA Insurance: rebuilt an internal form-builder with RxJS, taking new marketing form launches from weeks to days.

Earlier Roles

Software Engineer — DBM Apps, Singapore (Feb 2017 – Jun 2018)

Software Engineer — Validatis, Singapore (Oct 2016 – Jan 2017)

SPEAKING & COMMUNITY

- Speaker, *Web Fundamentals for Performant Web Apps*, JavaScript Developer Conference (JSDC) Taiwan, 2024
- Talks on RxJS, Vue.js, React Hooks and Docker at Singapore developer meetups
- Teaching Assistant, *Practical Nuxt.js Workshop*, JSConf.Asia, 2019

EDUCATION & CERTIFICATION

B.Eng. (Honours), Information Systems Technology and Design — Singapore University of Technology and Design. Dual focus: Artificial Intelligence and Business Analytics.

Agile Fundamentals Certified